

StatAna M1-M2 Reviewer

CS0073 Statistical Analysis and Modeling. Modules 1 and 2 of the midterm. Memory Aid boxes are study cues, not slide text.

Part 1: Module 1, Foundations and Descriptive Analytics

Data Value Chain

A value chain turns inputs into valuable outputs. The slide's analogy moves from coffee beans (raw material, storage), to instant coffee mix (processed material, inventory), to a coffee beverage (sales). Data creates value when insights lead to implemented decisions.

1. **Transactional Data:** raw records of events, such as purchases, claims, transfers, or payments.
2. **Database:** store, access, query, join tables, and aggregate data. This is sourcing data through analytic queries.
3. **Analytics:** describe distributions and trends, examine correlations and relationships, and measure model strength.
4. **Application:** test outcomes from varied inputs, scenarios, and sensitivity analysis; put analysis within reach of users.
5. **Decision:** use the application to act on insights and shorten repetitive tasks. Implemented action realizes data value.

Memory Aid

T-D-A-A-D: Transactional data, Database, Analytics, Application, Decision. Raw coffee becomes a drink; raw data becomes an action.

Five Data Sources

1. **Transactional data:** structured details of a transaction, such as a purchase or payment.
2. **Contractual, subscription, or account data:** product type together with customer characteristics.
3. **Surveys:** questionnaires that collect sociodemographic and behavioral information from a group.
4. **Data poolers:** companies that collect data for a purpose and sell it to enrich another source.
5. **Unstructured data:** material outside a traditional row-column database, including media, audio, sensor data, text, email, web pages, and PDFs.

Memory Aid

T-C-S-P-U: Transactions, Contracts/accounts, Surveys, Poolers, Unstructured material.

Gartner Analytic Ascendancy Model

The model moves toward greater value and difficulty. Descriptive gives hindsight and information; diagnostic gives insight; predictive gives foresight; prescriptive adds optimization.

Type	Question	Role
Descriptive	What happened?	Hindsight; information
Diagnostic	Why did it happen?	Insight
Predictive	What will happen?	Foresight
Prescriptive	How can we make it happen?	Optimization

Memory Aid

What, Why, Will, How: Descriptive, Diagnostic, Predictive, Prescriptive. Value and difficulty rise together.

Descriptive Analytics

Descriptive analytics asks what happened or is happening. Its two primary techniques are **data aggregation** and **data presentation**.

Branch	Measures Or Output
Numerical summaries	Totals and ratios
Central tendency	Mean, median, mode
Variation	Range, variance, standard deviation
Shape	Skewness, kurtosis
Position	Percentiles, quantiles, quartiles
Tabulation and visualization	Tables and charts

Memory Aid

C-V-S-P: Center, Variation, Shape, Position. Center = mean/median/mode; variation = range/variance/standard deviation; shape = skewness/kurtosis; position = percentiles/quantiles/quartiles.

Watch Out

The Excel section on slide p21 belongs to **Module 1 Subtopic 2, Descriptive Measures in Excel** (slide says "Module 2 Submodule 2").

Excel Descriptive Functions

Function	What It Does
SUM(range)	Adds numbers.
SUMIF(range, criteria, sum_range)	Adds values meeting one criterion.
AVERAGE(range)	Finds the arithmetic mean.
AVERAGEIF(range, criteria, average_range)	Averages values meeting one criterion. (slide says AVERAGE(range,criteria,average_range))
MEDIAN(range)	Returns the middle value.
MIN(range) / MAX(range)	Returns the smallest / largest value.
SMALL(range, k) / LARGE(range, k)	Returns the kth smallest / largest value.

COUNT(range)	Counts cells containing numbers.
COUNTA(range)	Counts nonblank cells.
COUNTBLANK(range)	Counts blank cells.
COUNTIF(range, criterion)	Counts cells meeting one criterion.
COUNTIFS(...)	Counts cells meeting multiple criteria.
SUMIFS(...)	Sums values meeting multiple criteria.
AVERAGEIFS(...)	Averages values meeting multiple criteria.
VAR(range) / VARP(range)	Sample / population variance; square of standard deviation.
STDEV(range) / STDEVP(range)	Sample / population standard deviation. (slide says STEVP)
QUARTILE(range, quart)	Returns a quartile, including Q1 and Q3.
PERCENTILE(range, k)	Returns a value at a specified percentile.

Memory Aid

Count, conditional, location, spread, position: COUNT-family; IF/IFS; AVERAGE/MEDIAN/MIN/MAX; VAR/STDEV; QUARTILE/PERCENTILE/SMALL/LARGE.

Watch Out

COUNT counts numbers; **COUNTA** counts nonblank cells; **COUNTBLANK** counts blank cells.

Watch Out

VAR and **STDEV** use a sample; **VARP** and **STDEVP** use a population.

Excel Slide Examples And Results

Slide	Example And Result
p25	Enrollment: COUNT(student numbers) = 72; COUNTA(exam scores) = 69 took the exam; COUNTBLANK = 3 absent.
p26	On-time performance: COUNT = 16; AVERAGE = 96.9%; VAR = 4.8; STDEV = 2.1.
p27	Order status: Complete COUNTIF/COUNTIFS = 4, SUMIFS total = 366.50; Pending count = 2, shown total = 366.50; Complete OR Pending count = 6, shown total = 527.00. The displayed subtotals conflict with the displayed combined total.
p28	AVERAGEIFS telephone expense: North = 275; South = 200.
p29	Sales-rep MIN/MAX: January 2,300/3,800; February 2,200/3,600.
p30	MEDIAN of EV/EBITDA multiples 2.5, 0.5, 1.4, 8.7, 2.7, 3.6 = 2.6.
p31	QUARTILE real-estate prices: Q1 = 387,000; Q3 = 639,000; IQR = 252,000; bounds 9,000 and 1,017,000; 5,500,000 and 1,095,000 flagged.
p32	Age percentiles (10th to 100th): 25, 30, 34, 39, 42, 48, 52, 56, 60, 65. At the 10th percentile, 10% are below age 25.
p33	LARGE(Table[Speed], k): k = 1 gives 120; 2 gives 100; 5 gives 90; 10 gives 78.
p34	SMALL(Table[Time], k): fastest 0:57:15; second 1:10:33; third 1:21:35.
p35	Pivot table: rows Gender; columns Region; filter Paid With; values Sum of Total Cost.

Memory Aid

Slide sequence: attendance, on-time performance, order status, telephone expense, monthly sales, median multiples, quartile outliers, age percentiles, speed rank, time rank, pivot table.

Watch Out

Median resists extreme values better than mean. The EV/EBITDA slide selects median for that reason.

Quartiles, Percentiles, and Outliers

Q1 marks the value below which 25% of observations lie; Q3 marks 75%. For the slide's prices, **$IQR = Q3 - Q1 = 639,000 - 387,000 = 252,000$** . Multiply IQR by 1.5 to get 378,000. Lower bound = $387,000 - 378,000 = 9,000$. Upper bound = $639,000 + 378,000 = 1,017,000$. The prices **1,095,000** and **5,500,000** exceed the upper bound.

Watch Out

Outlier fences use $1.5 \times IQR$, then subtract from Q1 or add to Q3.

A percentile is a position in an ordered distribution. The age slide divides people into ten equal groups: its 10th percentile is age 25, meaning 10% are below age 25.

Descriptive Tools and Process Control

1. **Pivot table:** organize and summarize data by row, column, filter, and value fields. The slide uses Gender, Region, Paid With, and Sum of Total Cost.
2. **Moving average:** average observations over a chosen number of past periods to smooth noise and reveal trends; the slide compares actual and forecast lines.
3. **Pareto chart:** bars from highest to lowest frequency or cost, with a cumulative percentage line, to prioritize which problem to address first.
4. **Statistical process control:** monitor variation to find special causes and signal corrective action. All processes vary; the goal is to reduce variation.

Memory Aid

P-M-P-S: Pivot summarizes, Moving average smooths, Pareto prioritizes, SPC monitors.

A control chart has a **center line at the average** and **upper and lower control limits** based on variability, set at **3 sigma** by default. Points within limits show common cause variation natural to an in-control process; points outside show special cause variation, often from external sources, and an out-of-control process.

Watch Out

Common cause = in control; special cause = not in control.

Part 2: Module 2, Diagnostic Analytics

Diagnostic analytics asks **why it happened** using historical data. It investigates anomalies, relationships, and underlying drivers.

Descriptive Analytics	Diagnostic Analytics
Uses historical data	Uses historical data
Reconfigures data into an easily read format	Identifies data anomalies
Describes current performance or outcome	Highlights trends and relationships
Learns from the past	Investigates underlying drivers
Answers what	Answers why

Three Functions

1. **Identify anomalies:** notice unusual results.
2. **Drill into analytics:** investigate details to discover an explanation.
3. **Determine causal relationships:** examine possible causes. (slide says "casual relationships")

Memory Aid

I-D-C: Identify anomalies, Drill into analytics, determine Causal relationships.

Question-To-Test Map

Question	Slide's Method
Which factors move together?	Correlation coefficient
Are categorical distributions different?	Chi-Square
Are two populations similar?	ANOVA (F-test); analysis of means (Z-test, T-test)

Memory Aid

Move, Distribution, Population: Correlation; Chi-Square; ANOVA/F-test or Z-test/T-test.

The objectives list Chi-Square, T-test, and one-way ANOVA, but these slides give no worked examples.

Association and Correlation

Association is any general relationship between two variables, including a curved one.

Correlation measures a linear relationship and may be positive, negative, or absent. A U-shaped pattern can be associated without a linear correlation.

Watch Out

Association is not always linear; correlation refers to a linear relationship.

One-Page Cram Sheet

Topic	Recall
Value chain	Transactional data > Database > Analytics > Application > Decision; value arrives when action is implemented.
Five sources	Transactions, contracts/accounts, surveys, data poolers, unstructured data.
Gartner	Descriptive what/hindsight; diagnostic why/insight; predictive will/foresight; prescriptive how/optimization. Value and difficulty increase.
Descriptive	Aggregate and present. Tree: central tendency, variation, shape, position; plus totals, ratios, tabulation, visualization.
Excel traps	COUNT numbers, COUNTA nonblank, COUNTBLANK blank; VAR/STDEV sample, VARP/STDEVP population; AVERAGEIF spelling.
Middle and rank	MEDIAN resists extremes; SMALL/LARGE return kth item; percentiles locate proportions.
IQR fences	Q1 387,000; Q3 639,000; IQR 252,000; 1.5 x IQR 378,000; bounds 9,000 and 1,017,000.
Other tools	Pivot summarizes; moving average smooths; Pareto prioritizes; control chart uses average center and 3-sigma limits.
Variation	Common cause within limits = in control; special cause outside limits = not in control.
Diagnostic	Why? Identify anomalies, drill into data, determine causal relationships.
Question > test	Move together > correlation; distributions > Chi-Square; populations > ANOVA/F or Z/T.
Relationship	Association can be curved; correlation is linear.